

ANSWERS

1.

Question	Answer	Marks
4(a)	$(\lambda =) v/f$ or $3.0 \times 10^8 / 4.7 \times 10^{14}$	C1
	6.4×10^N (m) or $3 / 4.7 \times 10^6$	C1
	6.4×10^{-7} (m)	A1
4(b)(i)	speed: decreases frequency: stays the same wavelength: decreases	B2
4(b)(ii)	$i = 45^\circ$ and $r = 30^\circ$	C1
	$(n =) \sin(i) / \sin(r)$ or $\sin(45^\circ) / \sin(30^\circ)$	C1
	1.4	A1
4(b)(iii)	straight continuation of ray in block and refraction away from normal in air	B1
	emergent ray parallel to initial incident ray	B1

2.

Question	Answer	Marks
7(a)(i)	X-rays and gamma (radiation / rays)	B1
7(a)(ii)	security marking / detecting counterfeit bank notes / sterilising water / fluorescent effects (e.g. fingerprints, blood, biological fluids) / (sun) tanning / sun beds / cancer treatment / curing inks and resins (e.g. in dental fillings) / production of vitamin (D) / generating electricity / solar panels / disinfection / sterilisation	B1
7(a)(iii)	(skin) cancer / cataracts / burns or destroys the retina or cornea	B1
7(b)(i)	critical angle	B1
7(b)(ii)	$(n =) \sin i / \sin r$ or $1 / \sin c$	C1
	1.6	A1

3.

Question	Answer	Marks
6(a)(i)	object within focal length / same distance from lens as the focal length	B1
6(a)(ii)	from divergent / parallel rays or rays followed back (behind lens meet at point on image or rays do not meet on a screen / do not converge or rays appear to come from image	B1
6(b)(i)	ray through optical centre from top of object	B1
	ray parallel to axis from top of object to lens and line from junction with lens to top of image	B1
	both rays extended back to image and ray parallel to axis correctly refracted through lens to appear to come from top of image	B1
6(b)(ii)	4(.0 cm)	B1

4.

Question	Answer	Marks
8(a)(i)	vertical biconcave lens shape along P	B1
8(a)(ii)	on right of P, middle ray continues in straight line	B1
	other two rays diverge from middle ray	B1
8(b)(i)	continuations of the three rays meet at a point and on continuation of middle ray	B1
	meeting point within eye and rays strike back of eye at different places	B1
8(b)(ii)	image is sharp / focused / not blurred	B1
8(b)(iii)	light diverges before entering eye or meeting point now on the back of the eye	B1
8(c)(i)	one point on principal axis 4.0 cm from the centre of the lens indicated	B1
	any two rays from: paraxial ray from tip of object to lens and seems to come from left-hand principal focus after refraction ray from tip of object towards right-hand principal focus (until lens) and paraxial after refraction ray from tip of object <u>reaching</u> the centre of the lens	B2
	image marked at correct intersection	B1
8(c)(ii)	virtual and light only seems to come from the image or light does not pass through image	B1
8(c)(iii)	E marked on the right of the dashed line	B1
8(c)(iv)	height of candidate's I / 3.5	C1
	$0.30 \leq \text{magnification} \leq 0.50$	A1

5.

Question	Answer	Marks
4(a)	clear angle is to / from the normal or 90° – angle between ray and surface	C1
	angle between/with the refracted <u>ray</u> / <u>beam</u> and the normal or vv or angle from the normal to the refracted <u>ray</u> / <u>beam</u> or vv	A1
4(b)	(refractive index =) $\sin i / \sin r$ algebraic or numerical in any form	M1
	1.5	A1
4(c)	refractive index smaller (for red)	B1
	larger angle of refraction (for red) or (red) bends / refracts less (towards normal)	B1
	assume comparison is for light entering glass at equal incident angles unless clear otherwise	
4(d)	P infrared and Q ultraviolet	B1

6.

Question	Answer	Marks
4(a)(i)	correct <i>i</i> and <i>r</i> marked	B1
4(a)(ii)	reflection shown at approximately the correct angle or ray along surface	B1
4(a)(iii)	correct refraction as red light enters prism showing less refraction than blue light	B1
	correct refraction as red light emerges from left side of prism	B1

4(b)(i)	<u>angle of incidence</u>	B1
	<u>smallest angle</u> when ray totally internally reflects or (when ray is) just internally reflected or when ray (refracted/emerges) along surface / at 90° to normal / angle of refraction is 90° OR angle of refraction when the angle of incidence is 90° B2	B1
4(b)(ii)	$\sin c = 1 / n$ in any form numerical or algebraic	B1
	42°	B1

7.

Question	Answer	Marks
3(a)(i)	it / light refracts / bends / moves towards normal or angle of incidence greater than angle of refraction	B1
	change in direction of the ray / refraction shows a change in speed (at the boundary) or $\sin(i) / \sin(r)$ is ratio of speeds or top of ray slows down first and enters the block first	B1
3(a)(ii)	(wavelength) decreases	B1
	(frequency) does not change.	B1
3(b)(i)	$(\sin(r) =) \sin(i) / n$ or $\sin(45^\circ) / 1.6$ or $0.71 / 1.6$ or 0.44 or $(r =) 21^\circ$	C1
	26°	C1
	64°	A1
3(b)(ii)	64° / angle of incidence / i is greater than the critical angle / 39° or angle of incidence = 64°	B1
	it undergoes total internal reflection	B1

8.

10(b)(i)	four colours from violet, indigo, blue, green, yellow, orange, red	B1
	correct order of four of the above colours	B1
10(b)(ii)	prism	B1
	slit (to produce narrow beam)	B1
	dispersion shown at first or second face, i.e. a ray becoming at least two rays at least one face	B1
	correct refraction on first and second faces for all rays shown	B1

9.

Question	Answer	Marks
5(a)	convex or converging	B1
5(b)	as it enters – refracts (towards normal) or changes speed	B1
	as it leaves, one of: • bends / refracts away from normal • speed increases • wavelength increases • emerges parallel to the (principal) axis	B1
5(c)	either emergent ray R or the ray through F extended to the left of the lens	B1
	both ray through F and R shown and image marked at intersection of extended rays	B1
5(d)	upright and virtual	B1

10.

Question	Answer	Marks
8(a)(i)	$n = \sin i / \sin r$ or $1.4 = \sin 55^\circ / \sin r$ or $(r =) \sin^{-1}(\sin i / n)$ or $(r =) \sin^{-1}(\sin 55^\circ / 1.4)$	C1
	36°	A1
8(a)(ii)	towards (the normal) and speed decreases (i.e. fourth box only)	B1
8(a)(ii)	stays constant	B1
8(b)(i)	<u>distance</u> between (one) focal point / principal focus and (the centre of) lens	B1
8(b)(ii)	at least one correct focal point labelled / clear / used	B1
	any two from drawn: paraxial ray from tip of O to lens and refracted through <u>correct</u> far F ray from tip of O through optical centre and undeviated ray from tip of O through <u>correct</u> near F to lens and refracted paraxially	B2
	I marked at intersection of correct rays	B1
8(b)(ii)	$3.4 \text{ cm} \leq \text{distance} \leq 3.5 \text{ cm}$	B1
	(magnification =) candidate's distance / 9.0 or candidate's height / 3.0	C1
	$0.37 \leq \text{magnification} \leq 0.40$ c.a.o.	A1
8(c)	on next page	
8(c)	any three from: light (from subject / object) focussed / refracted by lens by moving / sliding / adjusting lens <u>real</u> image (of subject / object) formed on digital light detector / film	B3

11.

Question	Answer	Marks
4(a)(i)	it decreases	B1
4(a)(ii)	it does not change	B1
4(b)(i)	$(c =) \sin^{-1}(1 / n)$ or $\sin^{-1}(1 / 1.6)$ or $\sin^{-1}(0.625)$	C1
	39°	A1
4(b)(ii)	reflected ray at Z and $i = r$ and no refracted ray	B1
	ray strikes vertical face perpendicularly and undeviated in air	B1

12.

Question	Answer	Marks
4(a)(i)	60°	B1
4(a)(ii)	0°	B1
4(a)(iii)	back along the initial path	B1
4(b)	Any three of • same size (as object) • same distance (from mirror as object) • upright • virtual • laterally inverted	B3

13.

Question	Answer	Marks
6(a)	behind the mirror or on the opposite side (of mirror) to object	B1
	same distance as object (from mirror)	B1
6(b)(i)	image marked at correct position by eye	B1
6(b)(ii)	ray from tooth to mirror and ray from mirror to eye	M1
	angles of incidence and reflection equal (by eye)	A1
6(c)	virtual or same size (as object) or laterally inverted	B1

14.

Question	Answer	Marks
10(a)	$3.0 \times 10^8 \text{ m/s}$	B1
10(b)	red orange yellow green blue indigo violet (any order)	B1
10(c)(i)	(-).6.(0) cm	B1
10(c)(ii)	any two rays drawn from: paraxial ray that refracts and seems to come from F_1 ray through the optical centre of lens ray that aims for F_2 but refracts and emerges paraxially	B2
	rays traced back to point	B1
	(point) labelled I and rest of image drawn down to the principal axis	B1
10(c)(iii)	1.7–1.9 cm	B1
10(c)(iv)	candidate's 10(c)(iii) / 3.0 evaluated	B1
10(c)(v)	any two from: a real image can be projected on to a screen light actually passes through a real image on same side (of lens) as object or on opposite side of <u>mirror</u> to object	B2
10(c)(vi)	correction of short-sight / myopia	B1
10(d)	light travels more slowly in glass or light changes speed	B1
	one side / left-hand side of wavefront slows down first	B1
	wavelength decreases or wavefront travels a shorter distance in the same time	B1

15.

Question	Answer	Marks
3(a)	middle ray angle of incidence labelled c	B1
3(b)	angle of incidence (of right-hand ray only) greater than the critical angle	B1
3(c)(i)	$(n=) \sin i / \sin r$ algebraic or numerical	C1
	1.3	A1
3(c)(ii)	$\sin c = 1 / n$ algebraic or numerical	C1
	answer in range 49° to 50.3°	A1

16.

9(c)(i)	electromagnetic and transverse underlined	B1
9(c)(ii)	$n = \sin(i) / \sin(r)$ or $(r =) \sin^{-1}(\sin(i) / n)$ or $(r =) \sin^{-1}(\sin(60^\circ) / 1.6)$	C1
	33(°)	A1
9(c)(iii)	1 $n = 1 / \sin(c)$ or $(c =) \sin^{-1}(1 / n)$ or $\sin^{-1}(1 / 1.6)$	C1
	39°	A1
	2 total internal reflection	B1
	$\theta > c$ or $57^\circ > 39^\circ$ or angle of incidence greater than the critical angle	B1
	3 reflected ray at P (correct angle by eye)	B1
	light in air at 30° (by eye) to vertical surface	B1

17.

Question	Answer	Marks
6(a)	any two of - ray <u>through</u> middle of lens undeviated - ray parallel to axis passes <u>through</u> focus, 3 cm from lens - ray <u>through</u> focus on left of lens parallel to axis after lens	B2
6(b)	inverted or real	B1
6(c)	image size / object size or image distance / object distance numerical or algebraic	C1
	rays from lens intersect on diagram (for image) and 1.3–1.7	A1
6(d)	projector or photographic enlarger	B1

18.

Question	Answer	Marks
9(a)	true false false true all 4 correct 2 marks	B1
	3 correct 1 mark	B1
9(b)(i)	correct refraction towards normal on entering prism	B1
	splitting into colours occurs at first surface	B1
	red at top and blue / violet at bottom <u>inside</u> and outside prism if shown	B1
9(b)(ii)	(different colours have) different speeds in the glass or different refractive indices	B1
	different (angles of) <u>refraction</u> or <u>refract</u> at different angles	B1
9(c)(i)	$3(.0) \times 10^8$ m / s	B1
9(c)(ii)	$(t=) d / s$ in any form algebraic or numerical, e.g.	C1
	500 s	A1
9(d)(i)	$(I=) Q / t$ in any form algebraic or numerical, e.g. 180 / 2	C1
	1.5 A	A1
9(d)(ii)	1st column helium nucleus or 2 protons and 2 neutrons	B1
	2nd column positive and neutral / no / charge / none	B1
	3rd column paper / <u>thin</u> metal / skin / few cm air etc. and (thick) lead / thick metal / concrete / stone / earth OR never completely stopped / nothing	B1

19.

10(d)(i)	normal indicated and angle of incidence indicated	B1
10(d)(ii)	$n = \sin i / \sin r$ or $1.5 = \sin 57^\circ / \sin r$ or $(r =)\sin^{-1}(\sin 57^\circ / n)$ or $\sin^{-1}(\sin 57^\circ / 1.5)$	C1
	34°	A1
10(d)(iii)	1 no change	B1
	2 decreases and decreases	B1
	3	
10(d)(iv)	ray in glass between normal and continuation of the incident ray	B1
	ray in air between continuation of the refracted ray and side of prism	B1

20.

Question	Answer	Marks
9(a)(i)	ray from right-hand corner of mirror to eye	B1
	any incident and corresponding reflected ray correct by eye	B1
9(a)(ii)	normal drawn at any intersection of incident and reflected ray	C1
	both r and i labelled correctly with normal	A1
9(a)(iii)1	cannot be formed on a screen or nothing at the image (position)	B1
	rays do not come (all the way) from the image or rays only appear to come from image	B1
9(a)(iii)2	(same distance) behind the mirror or same size (as object) or upright / erect or laterally inverted	B1
9(b)(i)	reflection in mirror occurs at any angle or total internal reflection (TIR) only occurs for $i >$ critical angle or there is no critical angle for the mirror B1	any 2
	TIR occurs from dense to less dense medium or in the dense(r) medium or from glass to air or inside / does not escape glass or from slow to fast (media) or mirror reflection from air to glass B1	B2
	(mirror) reflection is not total, e.g., not all reflected or better quality of image or multiple images from a mirror B1	
9(b)(ii)	$(n=) 1 / \sin C$ or $1 / \sin 44$	C1
	1.4	A1
9(b)(iii)	$n = \sin i / \sin r$ in any form, e.g. $\sin r = \sin 50 / n$	C1
	32° – 33°	A1
9(b)(iv)	ANY 2 lines from - more data per second or per unit time - less decrease in strength / amplitude / attenuation - less heat / power produced / wasted - less need for repeating or amplification stations - less interference / noise - more secure / less chance of cross-talk - lighter / less heavy	B1 B1

21.

Question	Answer	Marks
4(a)(i)	image height / object height or image distance / object distance	B1
4(a)(ii)	rays go to / meet at image or can be formed on a screen	B1
4(b)(i)	horizontal ray continued to bottom of I	B1
4(b)(ii)	2.4 cm	B1
4(b)(iii)	any two other rays drawn, e.g. through centre of lens	B1

22.

Question	Answer	Marks
10(a)(i)	both reflected rays correct by eye	B1
	image in correct position shown by continuation of rays behind mirror	B1
10(a)(ii)	virtual or upright or laterally inverted or same size (as object)	B1
10(b)(i)	line (joining points)	C1
	line joining points / particles on wave with same phase or line joining points along a crest etc.	A1
10(b)(ii)1	correct angle to surface $\pm 7^\circ$ with correct orientation and similar wavelength (by eye)	B1
10(b)(ii)2	at least two lines at smaller angle to surface with correct orientation in slower medium	B1
	showing smaller and constant wavelength with wavefronts deviated in correct direction	B1
10(c)(i)	$(v =) f\lambda$ or $2(000) \times 16$	C1
	32 000 cm / s or 320 m / s	A1
10(c)(ii)1	a range with 15–25 Hz as the lowest frequency and 15–30 kHz as the highest	B1
10(c)(ii)2	1.6 cm (if highest frequency is 20 000 Hz)	B1
10(c)(iii)	tube or other method to produce (narrow) beam of sound on source and/or on detector	B1
	stated detector or stated reflector	B1
	detector moved to find maximum loudness or angles i and r measured with suitable experiment	B1

23.

(a) (i)	3.0×10^8 m/s	B1
(ii)	$(\lambda =)c / f$ or $3.0 \times 10^8 / 4.3 \times 10^{14}$ 7.0×10^{-7} m	C1 A1
(b) (i)	decreases	B1
(ii)	$\sin(i) = n \times \sin(r)$ or $1.5 \times \sin(30^\circ)$ or 0.75 49°	C1 A1
(iii)	41°	B1
(c) (i)	dispersion at both surfaces and refractions in correct direction violet/blue light below the red light shown	B1 B1

- (ii) spectrum **or** band of (continuous) colours **or** colours of rainbow
red, orange, yellow, green, blue, (indigo, violet) B1
B1
- (iii) 1 X marked above red B1
2 it is / black surfaces are good absorbers (of IR radiation) B1
- (d) intruder / human IR beam broken IR reflected B1
being emits IR
or
intruder warm **or** does not reach **or** change detected B1
IR detected detector
24.
(a) smallest angle for total internal reflection **or** angle for refraction along surface B1
angle of incidence in (optically) denser medium B1
- (b) vertical ray continues undeviated B1
second ray (60° to horizontal) refracts away from normal into the air B1
third ray reflects internally **and** $i = r$ by eye **not** if any refracted ray B1
25.
(a) (i) long-sight **or** far-sight **or** hypermetropia B1
(ii) rays do not come together (on back of eye) B1
or rays do not converge (on retina)
or it / the image is not formed on retina / back of eye
or it / the image is formed behind retina / back of eye
- (b) (i) lens between rays and eyeball **and** a converging lens shown B1
(ii) converging or convex B1

26.		
(a)	angle of incidence	B1
	smallest angle for light to be totally internally reflected	B1
	or largest angle (of incidence) for ray to be refracted / emerge	
	or when light emerges along surface	
	or when angle of refraction is 90°	
(b)	(i) $n = 1/\sin C$ algebraic or numerical	C1
	2.5 or 2.46 or 2.458(59)	A1
	(ii) <i>left hand diagram</i> ray refracts away from normal and emerges into air at bottom left surface	B1
	<i>right hand diagram</i> reflected horizontal ray (by eye)	B1
	<i>right hand diagram</i> rest of ray completely correct to emerge into air at top face without refraction (by eye)	B1
27.		
(a)	$(n =)\sin i / \sin r$ or $\sin 55^\circ / \sin 33^\circ$	M1
	1.5(040274)	A1
(b)	(i) angle of incidence greater than critical angle or denser to rarer medium	B1
	(ii) reflected ray in correct direction (by eye) to edge of block and no second TIR (ign marked values)	B1
28.		
(a)	distance from (optical) centre to focal point (principal focus)	B1
(b)	(i) both Fs correctly positioned at ± 1 mm	B1
	(ii) two of: paraxial ray to lens through focal point to image	
	ray through optical centre	
	ray through focal point and then paraxial to image (ign. arrows)	M2
	X at crossing point of rays	A1
	(iii) 3.4–3.8 cm	B1

- 29.
- (a) $\sin i / \sin r$ or $\sin 50 / \sin 30$ C1
1.5(321) A1
- (b) moving from more dense to less dense medium B1
or moving to lower refractive index (air)
angle of incidence is greater than critical angle B1
- (c) less heat loss / more efficient B1
less chance of hacking / more secure / less interference
less reduction in signal / less need for boosting / larger distances possible / thinner
or less bulky
- 30.
- (a) $3.0 \times 10^8 \text{ m/s}$ B1
- (b) (i) 1. decreases **cao** B1
2. no change **cao** B1
3. decreases **cao** B1
- (ii) 1. i correctly marked (to normal) B1
2. r correctly marked (to normal) B1
- (c) (i) $\sin i / \sin r = n$ or $\sin i / \sin r = 1.5$ C1
 $\sin 89 / \sin r = 1.5$ or $\sin 89 / 1.5$ or 0.67(0.666565) C1
42° or 41.8025° A1
- (ii) i equal to / close to 90° $\sin i / \sin 45$ $\sin^{-1}(1/n) / \sin^{-1}(1/1.5)$
and r less than 45° = 1.5 **and** 41.8° B1
- or i never bigger than $\sin i > 1$ or r not be more than c
89°/90° B1
- 31.
- (a) ultra violet **and** infra-red [B1]
- (b) blue refracts / bends / deviates more [B1]
blue slows more (than red when entering glass) or blue and red have different speeds
(from each other in glass) [B1]
blue and red have different refractive indices [B1]

- 32.**
- (a) (i) any single value between 0 and 5.6 cm or a range all of whose values are correct [B1]
- (ii) any value beyond 5.6 cm [B1]
- (b) (i) ray through optical centre undeviated [B1]
other ray correct through or to axis 2.8 cm ($\pm \frac{1}{2}$ small square) from lens [B1]
- (ii) lines drawn meet after 11 cm or rays do not meet (on page) or rays almost parallel [B1]
- (iii) inverted, magnified, real all 3 needed and none wrong [B1]
- 33.**
- (a) converging **or** convex [B1]
- (b) image height $\div$ object height [B1]
- (c) (i) line when extended back joins top of image with intersection of ray and lens [B1]
- (ii) 3.0 ± 0.1 cm ecf from diagram [B1]
- (iii) any two further lines from top of stamp that appear to come from the top of the image [B1]
- 34.**
- (a) (i) speed of sound is (much) less than the speed of light (accept quoted values) B1
- (ii) **measure** the time delay (between the lightning and thunder) B1
divide distance by time/delay B1
- (b) (i) 3.0×10^8 m/s B1
- (ii) $(\lambda =) c/f$ **or** $3.0 \times 10^8 / 7.5 \times 10^{14}$ C1
 4.0×10^{-7} m A1
- (iii) (in any order) blue, green, orange, red, yellow, (indigo), (violet) **or** VIBGYOR C1
violet, indigo, blue, green, yellow, orange, red A1

(c) (i)	correct angle clear/labelled r	B1
(ii)	mark/determine entrance and exit points (e.g. trace rays back to glass) join/draw line between entrance and exit points	B1 B1
(iii) 1.	$n = \sin i / \sin r$	B1
2.	1.5/1.51/1.506176 with no unit (not just 1.5 without working out)	B1
(iv)	correct direction of refraction at both faces completely correct (above blue)	M1 A1
35.		
(a)	correct normal by eye correct angle of incidence between candidate's normal and incident ray correct angle of refraction marked between candidate's normal and BC	B1 B1 B1
(b)	decrease / change in speed / wavelength	B1
(c)	$n = \sin i / \sin r$ seen in any form	C1
	($\sin r =$) $\sin 45^\circ / 1.5$ or 0.47(14) seen	C1
	28(.1)°	C1
(d)	refracts less at first face and on correct side of normal	B1
	refraction at second face away from normal so that red ray and blue ray are diverging	B1
(e) (i)	angle of incidence is 0 or ray along normal/perpendicular to glass	B1
(ii)	angle of incidence/ θ is larger than critical angle total internal reflection occurs	B1 B1
(iii)	reflected ray drawn correctly and emerging without refraction from block	B1
(iv)	(eventually) light emerges (into air at Q) or light refracts (out at Q) or (weak) refracted ray appears	B1
	light emerging at Q coloured in some way or correct description of movement of reflected ray (as θ decreases)	B1

36.		
(a)	critical angle	B1
(b) (i) light refracted out into air and bent away from normal (ignore reflected ray)		B1
(ii) correct internal reflection (by eye) and no refracted ray (not at 90°)		B1
(c) ($t =$) distance/speed in any form numerical or algebraic (e.g. d/s , s/v $10/2 \times 10^8$)		C1
2.5×10^{-10} s		A1
37.		
(a)	$(\lambda =)v/f$ or $2 \times 10^8/4.7 \times 10^{14}$	C1
4.3×10^{-7} m		A1
(b) raybox/light source/ laser and mirror		pin(s) and mirror
shine ray at mirror		B1
mark rays		B1
measure i and r and equal		B1
repeat		B1
or		place two pins
		two more pins in line with image
		measure i and r and equal
		repeat
(c) (i) 83°		B1
(ii) total internal reflection or TIR cao		B1
angle of incidence exceeds critical angle		B1
(d) (i) (at least) one ray from X to mirror		M1
(at least) two rays from X to mirror and correct reflections		A1
rays traced back to marked I or I marked in correct place (by eye)		B1
(ii) 0.19 m		B1
(iii) less/no light wasted or hall brighter		B1

38.

- (a) more telephone signals (at one time)
 OR great(er) bandwidth; more data (per sec); more signals
 OR faster data/information transfer
 OR less attenuation; less energy/power/signal loss;
 OR long(er) distance (before regeneration)
 OR (more) secure
 OR less noise/interference OR high(er) quality/clear(er) B1
- (b) (i) correct normal and angle marked B1
- (ii) total internal reflection B1
angle of incidence is larger than critical angle B1
- (c) $(n =) \sin i / \sin r$ in any form numerical or algebraic C1
 $35(.2644)^\circ$ **unit ° needed** A1

39.

- (a) (i) refraction B1
- (ii) $(n =) \sin i / \sin r$ C1
 $\sin 45^\circ / \sin 29^\circ$ C1
 1.4585 to more than 1 sig. fig. A1
- (iii) the angle of incidence/incident angle is greater than the critical angle B1
 total internal reflection occurs B1
- (iv) correct refraction at C with ray parallel to AB B1
 correct reflection (and correct refraction on other face i.e. downwards) B1
- (b) (i) Any TWO of:
 undeviated ray through centre of lens
 ray parallel to axis through point 3 cm from lens on right after lens M2
 ray through point 3 cm to left of lens parallel to axis after lens A1
 rays converge and vertical image drawn and labelled I
- (ii) 1.2 ± 0.2 cm B1
- (iii) 1. real image (can be) formed on screen; virtual image not found on screen;
 rays converge on real image; rays do not converge on virtual image;
 rays only appear/seem to come from a point on virtual image B1
2. place object within focal length; between lens and focal point/principal
 focus B1
 view from other side of lens; look through lens; image same side as/behind
 object B1

- 40.
- (a) (i) central ray undeviated emerging from lens M1
 two outer rays meet the central ray at a point inside the eye **and** carry on to strike the retina A1
- (ii) light (from a single point) is spread over an area (on the retina)
or rays do not meet at a point on the retina
or image formed/rays meet/principal focus off retina B1
- (b) (i) **any** diverging lens: biconcave, planoconcave, convexoconcave –
 i.e. lens **clearly** thinner at the centre B1
- (ii) **all** rays diverge B1
- 41.
- (a) one outer ray parallel to principal axis C1
 three rays parallel to the principal axis A1
- (b) (i) (speed) reduced **or** slows down B1
- (ii) (speed) returns to original value/ 3.0×10^8 m/s B1
- (c) (i) ($f =$) c/λ **or** $3.0 \times 10^8 / 6.0 \times 10^{-7}$ C1
 $5(.0) \times 10^{14}$ Hz A1
- (ii) no effect/unchanged/($f =$) $5(.0) \times 10^{14}$ Hz B1

42.		
(a)	suitable block (semicircular/rectangular/prism)	B1
	suitable source of rays (e.g. ray box; pins on incident ray; laser not torch)	
	must be labelled on diagram or clear in text	B1
	diagram showing incident ray in glass/perspex (no arrow needed)	
	and correct refraction out into air	B1
	adjustment of (angle of incidence of) ray until along surface/just no longer emerges	B1
	(measure) correct angle marked or described clearly or C marked on diagram	B1
(b) (i) converging or convex		B1
(ii)	ray from top through middle of lens undeviated	B1
	other ray from top of object to same position on film	M1
	correct image labelled/drawn/marked	A1
(iii)	ratio of size/height/length/distance of image to size/height/length/distance of object	B1
(iv)	0.4(0) (± 0.05) no ecf (iii)	B1
(v)	upside down // inverted // real // other side of lens to object // nearer lens than object	B1
(vi)	(otherwise) not focussed // to/adjust focus // to produce a clear/sharp image //	
	(otherwise) rays do not converge on film // to converge rays onto film //	B1
	image on film // object at different distance	C1
	(otherwise) image formed in front of film // object now further	A1
43.		
(a)	(i) one correctly reflected ray (by eye)	B1
	(ii) two reflected rays traced back to an image	B1
	(labelled) image in correct position (by eye)	B1
	(iii) any two of:	
	virtual	
	full size/mag = 1 or same distance from mirror as C	
	laterally inverted (ign upright)	
	dimmer	B2
(iv)	more comfortable/no neck strain/no need to look up/reflects to eyes	B1
(b) (i)	(c =) $3(.00) \times 10^8$ (m/s) or $3(.00) \times 10^5$ (km/s) or used in equation	B1
	(f =) c/λ or $(3.0 \times 10^8/\text{their stated value}/330)/4.0 \times 10^{-7}$	C1
	7.5×10^{14} Hz or correct answer from stated value (incl. unit)	
	or $8.2/8.25/8.3 \times 10^8$ Hz	A1

44.

- (a) (i) horizontal ray from Q to pool edge **and** on to P from corner
critical angle marked C **or** obvious B1
B1
- (ii) for $i = 90^\circ$ **or** horizontal ray B1
angle(in water) equals/cannot be less than critical/C B1
- (iii) ($n =$) $\sin i / \sin r$ **or** $1 / \sin C$ **or** $1/n = \sin C$ **or** $\sin 90^\circ / \sin 49^\circ$
or $1 / \sin 49^\circ$ B1
1.3(2501) B1
- (iv) decreases B1
- (b) (i) any **two** of:
real
less bright
further from lens
beyond $2f$ B2
- (ii) straight ray from R to **top** of image B1
- (iii) where ray crosses principal axis, vertical line (L **or** drawn lens) B1
- (iv) paraxial ray from R to lens refracted to top of image
or paraxial ray from lens to top of image, traced back to R M1
F marked A1
- (v) 1.6 – 1.9 cm **or** attempt to use $1/u + 1/v$ C1
19 – 23 cm (2 sig. fig. only) A1
- 45.
- (a) (i) smallest angle of incidence for total internal reflection
or greatest angle of incidence that allows refraction
or angle of incidence for (refracted) ray along surface/angle of refraction 90° B1
- (ii) correct angle marked to normal (by eye) B1
- (iii) ray along surface or reflected ray correct (by eye) or both rays B1
- (b) ray in air refracted away from normal B1
- (c) refractive index = $\sin i / \sin r$ algebraic or numerical e.g. $1.5 = \sin 50 / \sin r$ C1
 31° accept 30.71, 30.7 degree symbol required somewhere A1

46.		
(a)	(i) correct direction of refraction at both faces (not along normal) blue below red and blue and red diverge	B1 B1
	(ii) any two from orange, yellow, green	B1
(b)	(i) total internal reflection or angle of incidence greater than critical angle	B1
	(ii) all colours reflected at same angle or all have $i = r$	B1
47.		
(a)	(i) one ray from M correctly reflected – checked by eye two rays from M correctly reflected – checked by eye – and traced back to image	C1 A1
	(ii) image point clearly marked at intersection/correct place checked by eye	B1
(b)	0.34 m cao	B1
48.		
(a)	(i) correct ray	B1
	(ii) correct angle marked to normal	B1
	(iii) (the angle) between the incident ray and the normal (at the point of contact)	B1
(b)	correct ray from hat to eye	B1
	0.85–1.15 m	B1
49.		
(a)	red	B1
(b)	(i) equal to	B1
	(ii) less than	B1
(c)	two correct refractions on Fig. 6.2	M1
	no dispersion and ray ends close to P	A1

- 50.
- (a) reflections correct by eye B1
- (b) **all** the ray **reflects** back (into the denser medium/glass)
or reflection **and** no refraction/escape into air B1
- (c) more calls **or** greater bandwidth **or** more/faster data(/sec)/information **or** better quality **or** less power loss/energy loss/attenuation **or** greater distance (between repeaters) **or** harder to tap **or** less noise/interference B1
- (d) $f = v/\lambda$ in any form numerical or algebraic C1
 3.3×10^{14} Hz A1
- 51.
- (a) any ray from top of object correct through lens within 1mm of optical centre or F B1
other ray from same point correct through lens meeting 1st ray **and** none wrong B1
- (b) (i) image size/object size (accept image distance/object distance **or** v/u) B1
(ii) 0.55–0.65 **ecf diagram in (a) sizes or distances** B1
- (c) rays completed to retina but would meet behind retina B1
- 52.
- (a) (i) reflected ray correct by eye **and** normal B1
(ii) 40° B1
 40° **or** same as angle of incidence B1
- (b) **diagram** with object, mirror, image in approx. correct position B1
at least 1 ray drawn from object/ray-box correctly reflecting from mirror B1
at least 2 rays extrapolated back to image position B1
- OR(b) **diagram** with object, mirror, image in approx. correct position B1
OR B1
Use of search pin behind mirror shown/stated B1
no parallax used to locate image **or** described B1
(ignore arrows/do not insist on dotted lines)

- 53.
- (a) 46 (°) B1
- (b) angle of incidence B1
 when refracted ray is along surface
 or minimum angle of incidence for Total Internal Reflection B1
- (c) $\sin i / \sin r$ or $1 / \sin C$ B1
 $\sin 63 / \sin 40$ or $1 / \sin 46$ C1
 1.39 (accept 1.3860 – 1.3902) A1
- (d) correct reflected ray by eye B1
- 54.
- (a) (i) correct normal (by eye to centre of circle) M1
 angle between normal and ray 1 marked A1
- (b) ray 1 sensibly reflected **and no** refracted ray B1
 ray 2 bends upwards (ignore reflection) B1
 ray 3 undeviated (ignore all rays leaving bubble) B1
- (c) (i) $\sin i / \sin r$ or ratio of speed in air/vacuum to speed in medium B1
 (ignore real/apparent depth)
- (ii) 1.33 or 0.75
 (accept 1.326, 1.3, 0.754, 0.8 **not** 1.325, 1, 0.76) B1